UNIVERSITY OF MUMBAI

UGC Grant Minor Research Project Report

Location Based Alert System using Twitter Analytics

Submitted in partial fulfilment of the requirements of

UGC Grant Minor Research Project No. 463

by

Mrs. Lifna C. S

Dr. (Mrs.) M Vijayalakshmi

DEPARTMENT OF INFORMATION TECHNOLOGY

V.E.S INSTITUTE OF TECHNOLOGY

2016 - 2017

Dedication Sheet

"All Praise, Laud, and Honour to Jesus Christ for his Amazing Grace"

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

31/7/17.

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Mrs. Lifna C. S **Dr. (Mrs.) M. Viayalakshmi**

Date: 31st July 2017

Place : Chembur

University of Mumbai

Research Project No: 463

NAME OF THE RESEARCHER: Mrs. Lifna C.S.
LECTURE IN I.T.
AMOUNT SANCTIONED: Rs. 23,000/-

Ref No. APD/237/ of 2017
January, 2017

To,
The Principal,
VES Inst of Tech.
Chembur.
Mumbai

Sir/Madam,

With reference to the Research Proposal forwarded by you in respect of Shri/Smt/Kum Lifna C.S, I am directed to inform you that the said proposal has been considered by the University and the Committee has recommended research grant as quoted above to the researcher.

The sanctioned amount will be disbursed in two instalments. The first instalment of 50% of the sanctioned amount will be disbursed within 15 days on receiving undertaking and RTGS Form (copy enclosed) from the researcher.

The researcher is expected to spend 50% amount initially from his/her own resources to carry out the work. **Please note that 50% balance amount, out of sanctioned grant will be released on the submission of utilization certificate including bills/vouchers/receipts in original on or before 31st March, 2017 failing of which the advance amount granted will be recovered from researcher which please note.**

Further, I am to inform you that the researcher will have to utilize the sanctioned amount before 31st March, 2017 and submit original bills/vouchers of the expenditure along with Utilization Certificate (copy enclosed) duly certified by the Principal /Director /Head /Institute /University Department/College, to the A.P.D. Section, after verification through the Accounts Section, Gr. Floor, Room No. 13, Fort, Mumbai – 400 032.

The report of the research work carried out by the concerned researcher will have to be submitted to the University on or before 31st July, 2017.

The Principal/Head of the Institute are requested to inform the researcher accordingly and arrange to forward his/her undertaking along with RTGS Form immediately to enable this office to release first instalment of the research grant, accordingly.

Yours faithfully,

For REGISTRAR

ANNEXURE- I

MINOR RESEARCH PROJECT PROPOSAL

	PART - A : GENERAL INFORMATION		
1	Basic Subject Area of Research	:	**Data Analytics (Social Analytics)**
2	Title of the Proposed Proposal	:	**Location based Alert System using Twitter Analytics**
3	Name, Qualification and Designation of the Principal Investigator / Co-Investigator	:	Principal Investigator Name : Lifna C.S Qualification : ME(IT) Designation : Assistant Professor Co-Investigator Name : Dr. Vijayalakshmi M Qualification : Ph. D (CSE) Designation : Professor
4	Teaching and Research Experience of Principal Investigator	:	Teaching : 09 years Research : 01 year
5	Name and address of the Institution where the proposal will be executed	:	VES Institute of Technology Hashu advani Memorial Complex VES Campus II, Chembur - 400074
6	Whether the College / University is approved by the UGC	:	Yes
7	Details of facilities provided / to be made available at the College / University	:	Internet Access to Amazon Cloud Service
8	Have you ever applied before for Minor Research Project? If yes, give details	:	**NA**
9	Whether the Project or part of Project is approved by the University for the Doctoral Degrees If Yes, give details	:	**NA**
10	Details of the Research Project and research funding (Major /Minor) received in the past and/ongoing projects.	:	**NA**

	PART - B : PROJECT DETAILS

1	Details of the proposed project to be undertaken: (Attach additional pages if required)

Origin, Need and Objective of the Research Proposal:

Origin:
The digital footprint of the society has undergone an abrupt change due to the extensive use of social networking sites [1-6]. This scenario has opened up a new arena for the users to express their opinions about a wide range of topics. If used judiciously, Social Analytics [7-15] can even address many sensitive social issues such as violation of Human Rights (HR).

Need: To utilize Social Analytics for Social Benefit

Objective:
To design a Platform for Location based Alert System to aid Government bodies in taking corrective action against the violation of Human Rights.

Rationale for taking up the proposed project and its interdisciplinary relevance:

People are more interested in social presence ….

1. Delhi Zoo incident (23rd Sept 2014) video was first posted on Twitter, prior to any News media catch up with the detailed information [12].
2. Almost every News Channels have their Twitter Accounts to catch up in the race [13- 16].
3. GeoTwitter as a Distributed Sensor System for Situational Awareness[17,18]

Relevance of Research and development in the field:

1. Etlinger, Susan, and C. Li. "A framework for social analytics." *Altimeter Group. USA. Published on* 10 (2011): 2011.
2. Lau, Raymond YK, Chunping Li, and Stephen SY Liao. "Social analytics: Learning fuzzy product ontologies for aspect-oriented sentiment analysis."*Decision Support Systems* 65 (2014): 80-94.
3. Aggarwal, Charu C. "An introduction to social network data analytics." *Social network data analytics*. Springer US, 2011. 1-15.
4. Parsons, Todd, et al. "Social analytics system and method for analyzing conversations in social media." U.S. Patent No. 8,682,723. 25 Mar. 2014.
5. Ferguson, Rebecca, and Simon Buckingham Shum. "Social learning analytics: five approaches." *Proceedings of the 2nd international conference on learning analytics and knowledge*. ACM, 2012.
6. Nadeem, Dr. "Social customer relationship management (SCRM): how connecting social analytics to business analytics enhances customer care and loyalty?." *International journal of business and social science* 3.21 (2012).
7. Amer-Yahia, Sihem, et al. "MAQSA: a system for social analytics on news."*Proceedings of the 2012 ACM SIGMOD International Conference on Management of Data*. ACM, 2012.
8. Castillo, Carlos, et al. "Characterizing the life cycle of online news stories using social media reactions." *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing*. ACM, 2014.

9. Ciulla, Fabio, et al. "Beating the news using social media: the case study of American Idol." *EPJ Data Science* 1.1 (2012)
10. Bandari, Roja, Sitaram Asur, and Bernardo A. Huberman. "The pulse of news in social media: Forecasting popularity." *arXiv preprint arXiv:1202.0332*(2012).
11. Stieglitz, Stefan, and Linh Dang-Xuan. "Social media and political communication: a social media analytics framework." *Social Network Analysis and Mining* 3.4 (2013): 1277-1291.
12. Delhi Zoo Incident (2016 Aug 08) [Online] (http://bit.ly/1B9x2gJ).
13. Twitter Acoount, @BreakingNews (2016 Aug 08) [Online] http://bit.ly/2aFxPC1
14. Twitter Account for CNNBreakingNews : @cnnbrk (2016 Aug 08) [Online] Available: http://bit.ly/2b6Yubb
15. Twitter Account for BBC BreakingNews : @BBCBreaking (2016 Aug 08) [Online] Available : http://bit.ly/2aTLaYi
16. Twitter Account for Times of India : @timesofindia (2016 Aug 08) [Online] Available : http://bit.ly/2ayJ913
17. Crooks, Andrew, et al. "# Earthquake: Twitter as a distributed sensor system." *Transactions in GIS* 17.1 (2013): 124-147.
18. MacEachren, Alan M., et al. "Senseplace2: Geotwitter analytics support for situational awareness." *Visual Analytics Science and Technology (VAST), 2011 IEEE Conference on*. IEEE, 2011.

Work Plan (including Detailed Methodology and Time Schedule):

Detailed Methodology

1. Extract Social Media data, tweets (Twitter).
2. Preprocess the collected dataset to cleanse data and retrieve location.
    - Find the synonyms for the 13 domain keywords to identify labels for rules (Dictionary)
    - Enlist unique rules via Python Program
    - Check the correctness of the constructed domain dictionary.
    - Revitalize TWUARM (Temporal Weighted Utility Association Rule Mining) algorithm.
3. Derive Association Rules between real world events related to violation of Human Rights.
4. Identify the transitive relationship between events and create a repository.
5. Locate the current events and predict upcoming events in the locality and send notifications to the local and central Governing bodies to take corrective actions

Time Schedule: **1 year**

Expected Results, Conclusion and Future Scope:

Expected Results & Conclusion

Proof of concept with empirical evidence leading to publications in Peer-reviewed Journals.

Future scope

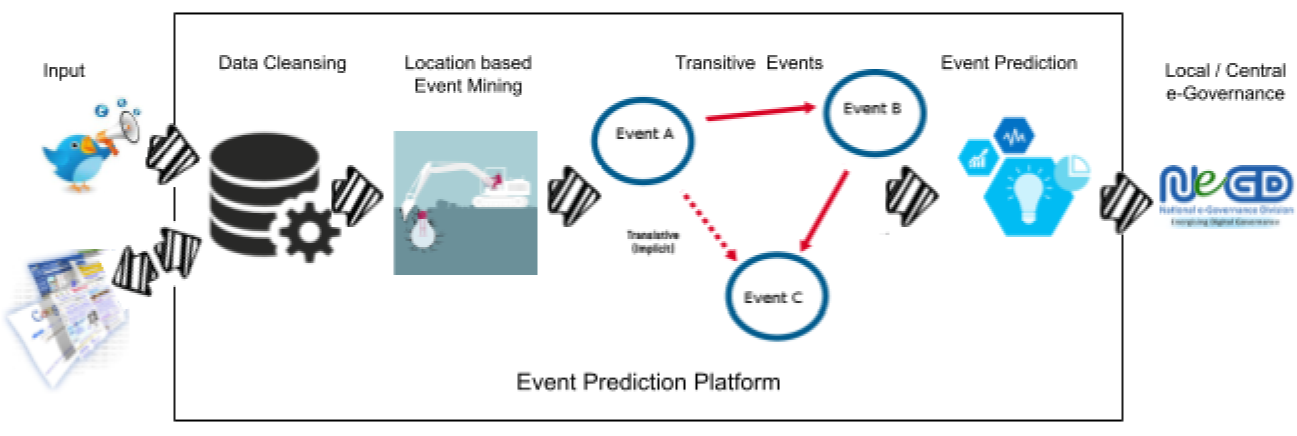
1. To incorporate other Social networking sites for data gathering and prediction.

	2. Build an ontology based domain dictionary.
2	Collaboration for the proposed project (if any) : **NA**
3	Details of financial requirements with justification

Sr. No.	Head	
1	Consumables and Chemicals	**2000/-** (Stationery)
2	Equipment (minor)	**15000/-** (Amazon Cloud Services)
3	Travel	**20000/-** (Presenting Paper in Conferences)
4	Books & Peripherals	**2000/-** (Access Research Papers + Journals)
5	Contingency	**1000/-**
	Total	**40000/-**

4	Any other information in support of the proposed project

<u>Proposed Architecture for Location based Alert system using Twitter Analytics</u>



Proposed Location based Alert System using Twitter Analytics

The working of the proposed system can be explained as follows:

1. Download tweets from Human Rights domain from Twitter
2. Extract location information while mining the tweets.
3. Derive Association Rules from the tweets reported and identify the real world events.
4. Create a repository by locating the transitive relationship between events.

5. Based on the transitive relationship between events, predict the upcoming events for the current reported event.
6. Notify the respective local and central Government bodies regarding the prediction to take corrective action.
7. Deploy the system on Cloud Platform for Real Time Analysis.

	PART - C : Bio-data and Endorsement
	Detailed Bio-data of the Principal Investigator as per Annexure -II
	Statement from the Present Employer as per Annexure-III

ANNEXURE - II

Detailed Bio-data

1. Name of the Applicant: Lifna C.S
2. Mailing Address: Computer Department,
   VES Institute of Technology,
   Chembur, Mumbai -400074
   a. Telephone: 9969268243
   b. Fax: 022 - 61532532
   c. E-mail: lifna.cs@ves.ac.in
3. Date of Birth: 27th November 1983
4. Educational Qualification (Starting from Graduation onwards):

Sr. No.	Degree	University	Year	Subjects	Percentage / CGPA
1	B.Tech	Cochin University of Science and Technology, Kerala	2005	Information Technology	75.47%
2	ME	University of Mumbai	2015	Information Technology	8.49 CGPA

5. A. Details of Professional Training and Research Experience, specifying period

   i. Teaching Experience: UG : 09 years
   PG : --
   ii. Research Experience : 01 year [CFDVS, IIT Bombay (2007-08)]

   B. Details of Employment (Past & Present)

   1. Worked as a Lecturer in Information Technology with KMEA Engineering College, Kerala from 3^{rd} Jan 2006 to 13^{th} Oct 2006.
   2. Worked as a Lecturer in Information Technology with K.J Somaiya Institute of Information Technology, Sion from 23^{rd} March 2007 to 31^{st} May 2007.

3. One Year experience as a Research Assistant in CFDVS, CSE Department of IIT Bombay under Prof. G. Sivakumar from 1st June 2007 to 25th June 2008.
4. Currently working as a Assistant Professor in Computer Department of Vivekanand Education Society's Institute of Technology, Chembur from 16th July 2008 – till date.

C. List of Significant Publications (Research Papers and books) during last five years. (06)

1. Lifna C.S, Dr. Vijayalakshmi .M, "Soft Grid – Big Data Analytics for Smart Grid", VESIT International Technological Conference – 2014 (I-TechCON 2014), January (03 - 04) 2014, IJAIEM ISSN 2319-4847.
2. Priya R.L, Lifna C.S, Dhanamma Jagli, Anooja Joy, "Rational Unified Treatment for Web Application Vulnerability Assessment", IEEE 2014 International Conference on Circuits, Systems, Communications and Information Technology Applications (CSCITA) , April 2014, DOI :10.1109/CSCITA.2014.6839283.
3. Rahul Vaswani, Rahul Vaswani, Manish Shahani, Lifna C.S, "Big Data Analytics over Online Transactional Dataset", IPASJ Int. Journal of Computer Science (IIJCS) ISSN 2321-5992, Vol.2, Issue 8, August '14.
4. Lifna. C.S, Dr. Vijayalakshmi .M, "Identifying Concept-Drift in Twitter Streams" , Elsevier Procedia Computer Science Volume 45, 2015, Pages 86–94, doi:10.1016/j.procs.2015.03.093, ICACTA 2015 held 26th - 27th March 2015 organized at D J Sanghvi College of Engineering, Vile Parle, Mumbai.
5. Sangeetha Rajesh, Lifna C.S, "UI Interface for Language Translator Module in Swasthya Slate (m-Health Tool)", IEEE 4th International Conference on Advances in Computing, Communications and Informatics (ICACCI), August 2015, 10.1109/ICACCI.2015.7275798
6. Vidya Rao, Pranoti Mane, Aditya Kumar, Lifna C.S, "Blind Aid : Travel Aid for Blind", IJCAx Journal October 2015, http://aircconline.com/ijcax/V2N4/2415ijcax02.pdf

6. Professional Recognition, Awards, Fellowships received: **NA**

7. Any other information: **NIL**

Place & Date: Chembur 31/7/17

Signature of the Applicant

Vivekanand Education Society's
Institute of Technology

V.E.S.
Since 1962

(Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)

Dr. (Mrs.) J. M. Nair
M. Tech., Ph.D. (IIT-B)
Principal

Ref. No. : VESIT/ MV/294/2016-2017

Date : 11/08/2016

ANNEXURE- III

Certified that

I. The University / College is approved under Section 2(f) and 12-B of the UGC Act.

II. The Institute welcomes participation of **Mrs. Lifna C.S** as the Principal Investigator in the Proposed Minor Research Project entitled **Location based Alert System using Twitter Analytics** And she will assume full responsibility for implementing the project.

III. The above research project / part of project is not funded by any other funding agency.

IV. The grant-in-aid received for the Research Project will be used to meet the expenditure of the project and the period for which the project has been sanctioned.

V. Institute undertakes the financial and other management responsibilities of the Project and undertake to submit Grant Utilization CErtificate and Project Report to the University.

VI. The Institution will provide in-house equipments and basic infrastructure and Other required facilities like administrative facilities to the Investigator.

V.E.S. INSTITUTE OF TECHNOLOGY CHEMBUR MUMBAI-74.

Seal of Institute

Signature

Head of the Institute

Principal
V.E.S. Institute of Technology
Hashu Advani Memorial Complex
Collector Colony, Chembur,
Mumbai-400 074. INDIA

UNDERTAKING TO BE SUBMITTED BY THE RESEARCHER

The Registrar,
University of Mumbai,
Fort, Mumbai - 400032

Subject - University Research Grant for the year 2016-17

Sir,

With reference to the subject mentioned above, I have to inform you that I am a permanent teacher of **Computer Engineering Department of VES Institute of Technology** College. I am submitting my undertaking as under :-

1. I shall utilize the entire research grant **(Rs. 23,000/-)** on my research project, which is sanctioned by the University.
2. I shall utilize 50% amount of the research grant initially from my own resources.
3. I shall submit to the University my statement of expenditure to the extent of entire research grant before 31st March 2017 along with original vouchers / bills, which will be duly sanctioned by the Principal / Director / Head of my College / Institute / University Department.
4. I shall not utilize the grant for purchase of any equipments or costly equipments such as Refrigerator, Computer / Laptop, Printer, Camera, Tape Recorder etc…
5. I shall not engage research assistant for my research work and incur expenditure on research assistant.
6. I shall return to my College / Institute / University Department, all the unused consumable or nonconsumable items purchased for my research work.
7. I also agree that the equipments / chemicals etc.. purchased for my research work shall not be claimed by me as my personal property.
8. I shall submit report of my research work to the Experts Committee in July 2017 and present my research work (Oral / Poster Presentation) as and when decided by the University.
9. I hereby declared that the above mentioned project has not been submitted by me for any other funding agency.
10. I further state that I did not receive Minor Research Grants from the University of Mumbai during the Financial Years 2014-15 and 2015-16.

Thanking You,

Yours sincerely,

Lifna

Signature of Research Teacher
Research Project Number : **463**

Mrs. Lifna C.S
604 / Staff Quarters, Hashu Advani Memorial Complex
VES Campus II, Collector Colony,Chembur Camp, Mumbai - 400074
Mobile No : 9969268243

Forwarded for necessary action through the Principal of College of Head / Director of University Department.

V.E.S. INSTITUTE OF TECHNOLOGY
CHEMBUR
MUMBAI-74.

Signature of the Principal of College
or Head/Director with seal of the Institute

Principal
V.E.S. Institute of Technology
Hashu Advani Memorial Complex,
Collector Colony, Chembur,
Mumbai-400 074. INDIA.

(6) Utilization certificate
(submitted: 31st March)

UTILIZATION CERTIFICATE

Certified that the sanctioned grant of Rs. 23,000/- **(Rupees Twenty three thousand only)** and received amount of first installment of **Rs. 11500/- (Rupees Eleven thousand five hundred only)** from the University of Mumbai, under the scheme of support for Minor Research Project entitled **"Location based Alert System using Twitter Analytics"** by **Mrs.Lifna C.S** has been fully untitled for the purpose for which it has sanctioned and in accordance with the terms and conditions laid down by the University.

Project No. **463**

University letter No. **APD / 237 / of 2017** dated: **January 2017**

Mobile No. **9969268243**

Principal Investigator. C.P.I Principal

Principal
V.E.S. Institute of Technology
Hashu Advani Memorial Complex,
Collector Colony, Chembur.

Date : 31st March 2017

College Seal

V.E.S. INSTITUTE OF TECHNOLOGY CHEMBUR MUMBAI-74

<u>**MINOR RESEARCH PROJECT**</u>

1. Summary of the Expenditure

Sr.	Bill No.	Date of Bill	Particular / Purpose	Amount (Rs)
1	VESIT Receipt No. 102836	6 Jan 2017	FDP - IT Innovations	2000.00
2	UTR No.. SBIN3170890 70133	30 Mar 2017	CDAC Short Term Course, "Text Analytics"	7500.00
3	HR-DEL2-133 194821-16387 41 (Amazon.in)	30 Mar 2017	Seven Layers of Social Media Analytics: Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data (Book)	2257.00
4	MH-QWCA-17 6059511-61197 (Amazon.in)	30 Mar 2017	Mastering Natural Language Processing with Python (Book)	799.00
5	(Amazon.in) Order No. 403-8492735-5 770700	30 Mar 2017	Cognitive Computing and Big Data Analytics (Book)	2430.00
6	MH-BOM4-100 4-2000195 (Amazon.in)	30 Mar 2017	WD 1TB External Hard Disk	4149.00
7	(Amazon.in) Order No. 403-4904698-9 020343	30 Mar 2017	4GB RAM	1894.00
8	Dhiraj Xerox Bill No.4396	31 Mar 2017	Printout paper + Stationery	407.00
			Total	21436.00

2. Stock Entry is required for purpose of various items.
3. Library Accession Number is required for purchase of Books.
4. Utilization Certificate is required.
5. Each Bill & Certificate is to be certified by the Principal alongwith the Stamp of the College.
6. Copy of the Sanction Letter is required.
7. Revenue Stamp Receipt above amount of Rs. 5000/-

V.E.S. INSTITUTE OF TECHNOLOGY CHEMBUR MUMBAI-74

HOD, IT.

Abstract

The digital footprint of the society has undergone an abrupt change due to the extensive use of social networking sites [1-6]. Analytics upon such social tools help in listening and analyzing customer interactions about products or services and address customer concerns promptly. In today's industry, enterprises are rigorously blending social intelligence with enterprise business intelligence to achieve competitive intelligence. So, the ongoing process of Social Analytics cannot be overlooked. The inferences driven from Social Analytics, are spontaneous and it varies drastically, leading to concept-drift. If used judiciously, Social Analytics [7-15] can even address many sensitive social issues such as violation of Human Rights (HR). Hence, there is a need to utilize Social Analytics for societial benefit.

The objective of the study was to design a platform for Location based Alert System which can aid Government bodies in taking corrective action against violation of Human Rights. Twitter being an active social tool, rendered exciting leads. The first phase of study was to establish, the temporal relationship between tweets from Human Rights domain. This domain was selected, as it deals with real-time social issues. The NGO from India - Shakti Vahini, which is an authorized organization was zeroed as an authenticated source. When tweets were examined, hidden contexts were deciphered via extensive preprocessing. Three mining techniques were experimented upon the refined dataset to study the confidentiality of the rules. Among them, the performance of the revised Temporal Weighted Association Rule Mining (TWUARM) was magnificent in that process. In this course of study, transitive property was also revealed. This transitive property was very significant to derive valuable inferences in Human Rights domain to predict events, and take corrective actions.

The next phase of study, was to locate events from tweets in the Indian context. The locations extracted from tweets were successfully plotted on to Indian Map. This visualization revealed the importance of integrating News Analytics with Social Analytics for deriving precise inferences about events. Work is in progress to tune the existing model towards predicting the upcoming events before integrating the findings with Local and Central Government bodies for taking corrective actions.

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Abbreviations and Nomenclature

ARM	Association Rule Mining
NLP	Natural Language Processing
NER	Named Entity Recognizer
SC	Support Confidence Framework
TWUARM	Temporal Weighted Utility Association Rule Mining
WUARM	Weighted Utility Association Rule Mining

Chapter – 1 : INTRODUCTION

The digital landscape of India is evolving quickly, with the extensive use of internet connectivity and smart phones. The plethora of activities, taking place in the social networking sites like Twitter, Facebook, LinkedIn and WhatsApp using these digital devices, generate a humongous amount of data about people preferences, behavior and sentiments. This deluge of data, derived from these social tools, is amenable to extract useful insights. This is the basic thrust towards Social Analytics.

1.1 Motivation

The inferences drawn from Social Analytics is relevant to, a wide range of stakeholders like marketers, sales persons, HR managers, product designers, investors and so on. The ability to connect and draw inferences makes Social Analytics, hugely exciting for anyone who is interested in understanding and influencing past, present and future trends. These results in usage proliferation contributes to better quality datasets, which leads to improved application of Big Data Analytics. In today's industry, enterprises are rigorously blending social intelligence with enterprise business intelligence to achieve competitive intelligence. So, the ongoing process of Social Analytics cannot be overlooked. If used judiciously, Social Analytics [7-15] can even address many sensitive social issues such as violation of Human Rights (HR). Hence, there is a need to utilize Social Analytics for societial benefit. This motivated us to select the respective topic.

1.2 Relevance of Location based Alert System

The digital footprint of the Indian society has undergone an abrupt change due to the extensive use of social networking sites [1-6]**.** This scenario has opened up a new arena for the users to express their opinions about a wide range of topics. People are more interested in social presence. Incidents such as Delhi Zoo incident happened on 23rd Sept. 2014, was first posted on Twitter prior to any News media catch up with the detailed information [12]**.** Even almost every News Channels have their Twitter Accounts to catch up in the race **[**13-16]**.** Also, social media tools such as GeoTwitter, a Distributed Sensor System for Situational Awareness [17,18] has changed society's perspective. Here comes the relevance of Location based Alert System using Social Analytics. Twitter being an active social tool, rendered exciting leads.

Violation against Human Rights has now become a common phenomenon and the concerned government authority's needs to be alerted about the upcoming consequences at an earlier stage. But, the government authorities are failing the extract meaning full information from these social media posts as these posts are concise. Our main objective was to address the Hidden Context challenge within a social media post and to utilize it for social benefit by selecting Human Rights as the application domain. The below mentioned scenario, clearly explains the motto behind selecting the Human Rights domain for the study. In the Human Rights domain, each tweet represents some real world event, and there also exists some temporal and transitive relationship between them. Therefore, the objective was to reveal the implicit transitive property that exists within the identified events. For example, Event A leads to Event B, and Event B in turn leads to Event C or Event D and so on. This implicitly reveals that, Event A leads to Event C or Event D represented as follows:

$$\left.\begin{array}{c} \text{Event A} \rightarrow \text{Event B} \\ \text{Event B} \rightarrow \text{Event C} \\ \text{OR} \\ \text{Event B} \rightarrow \text{Event D} \end{array}\right\} \quad \begin{array}{c} \text{Event A} \rightarrow \text{Event C} \\ \text{OR} \\ \text{Event A} \rightarrow \text{Event D} \end{array}$$

This can also be illustrated, with the help of the following example: Let Child Harassment be a known real world event, say A and Protest is another real world event say, B. This may lead to another Event C, as Criminal Arrested or Event D says, Child Rehabilitated. This clearly explains, the transitive property implied from Event A to Event C or Event D. Although there exists transitive property among these events, in some cases B, C or D may be unknown to the social media. In such cases, identification of unknown events turns out to be vital for discovering such temporal rules. Integrating these inferences with location could aid Local and Central Governing bodies towards improving the efficiency of e-Governance.

1.3 Objectives

The objective of the study was to design a platform for Location based Alert System, which could aid Government bodies in taking corrective action against violation of Human Rights. The study conducted over Twitter Data Streams was able to achieve the following objectives :

- Extract hidden information from tweets.
- Establish the presence of Temporal Association between Tweets.

- Locate events from tweets in Indian context.
- Visualize the findings using Heat maps on India Map.

1.4 Chapterization of Research Project Report

The report is mainly organized into five chapters. Chapter 1 covers, the motivation behind selecting the topic; relevance of Location based Alert System; justify the selection of Twitter for this study; listing the objectives set for the experimental study; and winds up with the observations and findings. A detailed literature review was carried out and summarized in Chapter 2. This Chapter, winds up with technical intricacies involved in processing Twitter Data Stream. The proposed methodologies adopted during the phases of experimental study is comprehended in Chapter 3. Chapter 4 illustrates, the implementation and proof-of-concept for these adopted methodologies. The findings of the experimental study with future scope is briefed in Chapter 5. Report concludes with a summary on the books, papers and websites referred.

1.5 Observations and Findings

The Twitter Data Stream was extensively studied to establish the presence of temporal rules. During the experimental study, it was observed that, enterprises are rigorously blending social intelligence with enterprise business intelligence to achieve competitive intelligence. So, the ongoing process of Social Analytics cannot be overlooked. The first phase of study was able to establish temporal relation between tweetsby normalizing the tweets gathered. Even the study was able to revitalized state-of-art Association Rule Mining, as Temporal Weighted Utility Association Rule Mining (TWUARM). In this course of study, transitive property was also revealed. This transitive property was very significant to derive valuable inferences in Human Rights domain to predict events, and take corrective actions. These were the parallel findings of the primary objective.

The next phase of study, was to locate events from tweets in the Indian context. The locations extracted from tweets were successfully plotted on to Indian Map. This visualization revealed the importance of integrating News Analytics with Social Analytics for deriving precise inferences about events. Work is in progress to tune the existing model towards predicting the upcoming events before integrating the findings with Local and Central Government bodies for taking corrective actions.

Chapter – 2 : LITERATURE REVIEW

This chapter introduces the background knowledge required for the topic with a discussion on the latest related work. Section 2.1 briefs the papers related to Twitter Data Streams. Section 2.2 discusses the stat-of-the-art Text Mining techniques. Section 2.3, 2.4 and 2.5 summarizes the related work in Association Rule Mining which helped to revamp Weighted Utility Association Rule Mining into Temporal Weighted Utiltiy Association Rule Mining (TWUARM). Section 2.6 concludes the literature survey by discussing the exixting methodologies used for creating Domain Dictionary.

2.1 Twitter Data Stream

Before diving deep into the current research work, it is imperative to have some knowledge of Twitter. Twitter (is undoubtedly one of the popular micro-blogging site (Chauhan, 2016; Ramm, Sherfesee, Shalhoup, & Dawson, 2010; Smith, 2016) (approx. 500 million tweets sent per day) where users search social-temporal information (Jack; Dorsey, Glass, Stone, & Williams, 2006b) such as breaking news, tweets about celebrities, trending topics etc... It is also used as a medium for real-time information dissemination during election campaigns. Within the Twitter Developers Community there exists many Real-Time Twitter Analytics & Visualization Tools [19] (Kilpatrick, 2015), which crawl through Twitter streams by aggregating, slicing, dicing and ranking the data to deliver some meaningful insights on Twitter activities and trends. Even-though these tools exist, they are all proprietary versions. Therefore, designing a system to extract events from social networking sites requires a thorough study upon the existing state-of-the-art techniques. These techniques were then fine-tuned for the specific purpose of establishing a temporal association between tweets in the Human Rights domain. The following subsections summarize the literature review performed for the same.

Papers [20-25] (Adedoyin-Olowe, Gaber, & Stahl, 2013; Althoff, Borth, Hees, & Dengel, 2013; Fiaidhi, Mohammed, Islam, Simon Fong, & Tai-hoon Kim, 2013; Lee et al., 2011; Mathioudakis & Koudas, 2010; Tare, Gohokar, Sable, Paratwar, & Wajgi, 2014) discussed the various supervised learning techniques adopted for classifying and forecasting the tweets in real-time. In papers [26,27] (Becker, Naaman, & Gravano, 2011; Psallidas, Becker, Naaman, Tech, & Gravano, 2013), Becker proposes an online clustering algorithm to classify tweets into known and unknown events. The attempt of this study was to combine her work with the transitive property that exists between these known and unknown real world events and come up with a novel initiative in social reform using the active social networking media.Rosa [28] (Rosa, Carvalho, & Batista, 2014) introduces the application of Fuzzy logic for classifying trending topics. In short, there is no existing foolproof state-of-the-art technique which can be applied for mining any Twitter Data Stream. Since the objective of the study is to discover temporal rules, the next focus of study is upon Association Rule Mining followed by a survey on Temporal Association rule Mining (TARM).

2.2 Association Rule Mining (ARM)

As mentioned in [29] (Jiawei & Micheline, 2006) Association Rule Mining (ARM) is considered as an important technique for the extracting hidden knowledge. Apriori was the first algorithm for mining association rules. In the recent years, many researches have proposed very efficient algorithms for mining frequent itemsets as discussed in [3] (Aggarwal, 2014; Ventura & Luna, 2016).

2.3 Temporal Association Rule Mining (TARM)

In paper [30] (Yin, Hsieh, Yang, & Hung, 2014), Yin discussed a GLF (Global-Local Frequent Pattern) Miner to generate global and local frequent patterns without prior domain knowledge efficiently. Sourabh Jain [31] (Jain, Jain, & Jain, 2013), reviewed the existing Fuzzy based TARM techniques and emphasized the importance of utility mining. In paper [32] (Shirsath & Verma, 2013) Pradnya presented a summary of different incremental TARM techniques. According to the author the performance of the PPM (Progressive Partition Miner) and SPF (Segmented Progressive Filter) is good.

Huang [33] (Junheng-huang & Wang-wei, 2007) presented a SPFA (Segmented Progressive Filter Algorithm) based on scan reduction technique to generate all candidate k-itemsets from candidate 2-itemsets. In paper [34] (Chin-Chen Chang, Yu-Chiang Li, & Jung-San Lee, 2005), Chang proposed New Fast UPdate Algorithm (NFUPA) for efficiently deriving incremental association rules from large transactional databases. The algorithm avoids re-scanning of original database, to detect new, frequent itemsets and delete infrequent itemsets invariably. Further, in-order to exclude least significant rules, the state-of-the-art techniques in the Weighted Utility Association Rule Mining (WUARM) were explored and summarized as follows.

2.4 Weighted Utility Association Rule Mining (WUARM)

Venu [35] (Kuthadi, 2013), introduced an enhanced Association Rule Mining with two additional calculations to validate the utility and its consistency in the mined association rules. The utility is validated by the integrated calculation of the cost/price efficiency of the itemsets and its frequency. The consistency validation is performed at every defined number of windows using the probability distribution function, assuming that the weights are normally distributed.

Dhanda [36] (Dhanda, 2011) proposed an innovative utility sentient approach for the mining of interesting association patterns from transaction database. First, frequent patterns are discovered from the transactional database using the Apriori algorithm. From the frequent patterns mined, this approach extracts novel, interesting association patterns with emphasis on the significance, quantity, profit and confidence.

In paper [37] (Sulaiman Khan, Muyeba, & Coenen, 2008), Khan introduced Weighted Utility Association Rule Mining (WUARM), which considers the varied significance of individual items as weights and different frequency values of those items as their utilities. Thus, weighted utility mining focuses on identifying the itemsets with weighted utilities, higher than the user specified weighted utility threshold.

All the above mentioned techniques were applied upon an incremental dataset and not on a temporal datasets. So, to discover temporal rules, the dataset was treated as temporal and WUARM was applied upon this dataset. To suit the mining requirements of the proposed system, WUARM was revitalized as Temporal Weighted Utility Association Rule Mining (TWUARM). In-order to apply TWUARM, on the tweets there is a need to define domain dictionary. The following subsection discusses the various methods used to create domain dictionaries for specific purposes.

2.5 Methodologies to create Domain Dictionary

In paper [38] (Lal & Asnani, 2014) Mily discusses a novel method to extract implicit aspects from the multi-aspect reviews with the help of NLTK WordNet [39] (Bird, Loper, & Klein, 2001). The proposal was to segment the multi-aspect review into multiple single aspects. Opinions from these aspects are then polled to determine positive and negative comments for efficient decision making. Falleri [40] (Falleri et al., 2010), introduced a fully automated system to comprehend software programs. Implicit concepts are extracted with the help of synonyms, hyponyms and hypernyms associated with explicit concepts from the defined NLTK WordNet. Both papers deal with taming implicit concepts using NLTK WordNet. But, NLTK WordNet cannot be applied in this context due to the inherent properties of tweets such as,

(1) Each tweet is restricted to 140 characters;

(2) Tweets are spontaneous;

(3) There does not exist a standard lingo for the tweets;

(4) Tweets are posted by each community in their agreed upon lingo which is sometimes difficult to decipher.

These properties impose hindrance in designing a fully automated system to create a domain dictionary for a social networking site like Twitter. Thus, the focus of this study is to, identify temporal association rules that exist between the tweets in the Human Rights domain. It was inferred that, only with the help of human intervention sub-domains can easily identified. This justifies how the 13 sub-domains covering all the major concepts of Human Rights were finalized.

2.6 Natural Language Processing (NLP)

As tweets were posted by individuals, the location information attached with tweets were of least significance compared to the actual location mentioned in the tweet text. So, a detailed study on Natural Language Processing was required to extract locations mentioned in the tweet before they could be attached with the tweets. Natural Language Processing [41] is one of the major components of Artificial Intelligence and Computational Linguistics. It gives computers the ability to understand human speech with the help of machine learning techniques.

Parts-of Speech (POS) tagging is one of the major tasks in NLP [41]. Among the various applications of POS tagging, Named Entity Reconizers (NER) has a major role is extracting named entities from a document. Various educational institutions and organizations are having specialized research centers developing fool-proof NERs and provide APIs for accessing them. Google, IIIT Hyderabad, Stanford University [42 - 45] are pioneers in this field.

Chapter – 3 : PROPOSED METHODOLOGIES

Chapter 3 discusses the procedures, techniques and methodologies developed and adopted during the phases of experimental study. Section 3.1 illustrates the architectural details involved in designing, a system to extract location based temporal rules. Section 3.2 describes the processes involved in data collection. Before mining the rules, data gahered has been normalized. Section 3.3 details the steps undergone towards data normalization. Section 3.4 gives the details of the proposed algorithm for Association Rule Mining, TWUARM. Section 3.5 mention the data visulization used to improve the readability of the derived results from Section 3.4. The chapter concludes with a detailed summary of the experimental setup of the study through section 3.6.

3.1 Proposed System Architecture

Figure 1, depicts the proposed architecture for Location based Alert System.

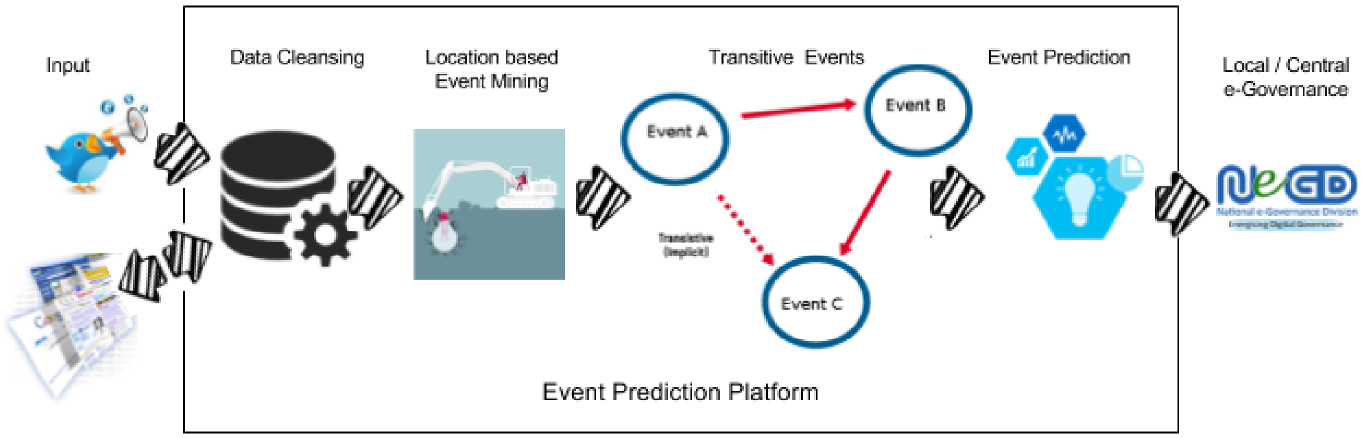


Figure 1 : Proposed Architecture for Location based Alert System using Twitter Analytics

No	Name of the NGO	Twitter Account Name	#Tweets
1	Shakti Vahini	@shaktivahini	2448
2	Human Rights Watch	@hrw	54.3K
3	UN Human Rights	@UNrightswire	7210
4	Human Rights First	@humanrights1st	14K
5	Human Rights at FCO	@FCOHumanRights	11.8K

TABLE 1 : TOP5 TWITTER ACCOUNTS UNDER HUMAN RIGHTS DOMAIN (Jack; Dorsey et al., 2006a)

To validate the proposed system, top 5 Twitter Accounts under the Human Rights domain were identified as given in Table 1. The proposed system was experimented to mine user timeline tweets posted on active Twitter Accounts of NGO's, Shakti Vahini (India) and Human Rights Watch (Global).

3.2 Data Collection

For developing any system, deluge of data is required. But, the inherent properties of tweets as mentioned earlier, turned out to be a hurdle while gathering data from user timeline tweets. Thus, the data gathering procedure was extensively carried out in three steps to extract the maximum data from the tweets as discussed in the following sub-sections.

3.2.1 Downloading Tweets

The first step in data collection was to create a Twitter application via a valid Twitter account. Then, user timeline tweets from @shaktivahini and @hrw were downloaded using tweepy (Jack Dorsey, Glass, Stone, & Williams, 2006; Roesslein, 2009) module in Python. But, the numbers of tweets downloaded were limited to the recent 3200 tweets/call. To overcome this challenge, tweets were downloaded on a daily basis. The downloaded tweets were then merged by discarding all the repeated tweets to form the raw dataset. Refer Appendix A and Appendix B for the sample raw user timeline tweets downloaded from @shaktivahini and @hrw.

3.2.2 Expanding Tiny-URLs in Tweets

The basic characteristic of any tweet, as defined by Twitter is to limit the text length to 140 characters. To help the users to overcome this limitation, Twitter has created its own Lingo (Beal, 2010; Jack et al., 2013). Twitter also allows URLs to be included in the tweets for effectively expressing one's view. As the characters of a tweet are restricted to 140, even the appended URLs are compressed into tiny URLs (of 22 characters) using the t.co (Jack et al., 2011) link service. Any tweet in this abbreviated form turns out to be meaningless for mining. To improve the effectiveness of such tweets, Becker [26,27] (Psallidas et al., 2013) has introduced the concept of expanding such tiny URLs into its original form for meaningful parsing. For this purpose, urlclean and Beautiful Soup modules defined in Python were used. Expanding such tiny URLs could help in revealing some unknown real world events that are hidden in these tweets.

3.2.3 Crawling webpages of Expanded URLs

As an extended attempt to improve the effectiveness of the tweets, web pages of the expanded URLs were crawled for further Text Mining. By this novel method, a tweet is stretched to its maximum, to extract the concepts contained in it. Such extracted hidden concepts, were helpful in depicting the link between the events referred in the tweets. And thereby, establish the temporal transitive property.

3.3 Data Normalization

During the first step, a tiny tweet of 140 characters was extended to around one lakh character by crawling the webpages of expanded tiny-URLs. Next step was to cleanse, this deluge of data for mining. Cleansing this raw data obtained after crawling the web pages, turned out to be a Herculean task. Therefore, the data preprocessing was performed in six stages.

In the first stage, HTML tags were removed using functions defined in Beautiful Soup [46] (Richardson, 2004). During this step, locations mentioned in the tweets were extracted using PyNER interface for Standford Named Entity Recognizer. After removing the HTML tags, in the second stage special characters and numbers need to be eliminated. For this, a regular expression was written in Python and a simple text replacement function was invoked. In the third stage, meaningless words from the gathered data were removed using PyEnchant [47] (Kelly, 2011) module. Two methods were applied in the fourth stage of data preprocessing, that is, removal of stop-words. (1) NLTK Stop-word Corpus [39] (Bird et al., 2001) was used to remove general stop-words appearing in WordNet; (2) NGO-Account related stop-words were eliminated by manually creating a dictionary for this special purpose.

The fifth stage of cleansing was to find the root words for each extracted word appearing in the tweet by stemming them. For this purpose, various WordNet Stemmer's such as Port Stemmer, Lancaster, Snowball and WordNet Lemmatizer [48] (Miner, 2015) were experimented upon. Among these, WordNet Lemmatizer gave most desirable result. The reason for better results was that, WordNet Lemmatizer was using the WordNet's built-in Morphy function, to return the input unchanged, if the root word turns out to be meaningless. All the above steps were performed on the original tweet, expanded tiny-URLs and their associated crawled webpages. At the final stage of data preprocessing, these three cleansed data components for each respective tweet were merged together, to form a single Tweet Corpus.

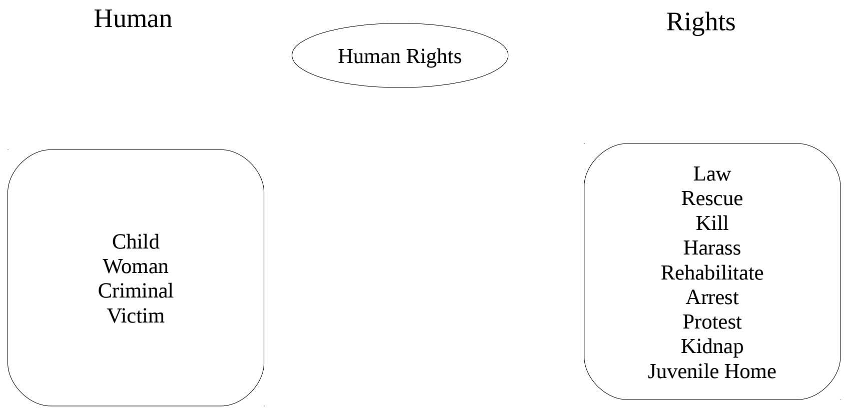


Figure 2. Sub-domains under Human Rights

In order to perform Association Rule Mining upon the cleansed dataset, grouping the stem words into the identified sub-domains was an inevitable step. The stem words in each of the merged Tweet Corpus obtained after preprocessing was finally grouped into 13 sub-domains as mentioned in Figure 2. For example, stem words trafficker, abuser, kidnapper etc... appearing in the group, 'Criminal' was replaced by the Group name, 'criminal'. As the tweets were context based, human intervention was required.

3.4 Mining Temporal Association Rules

Upon this refined data-set, an experiment was conducted using three mining techniques, to study the utility of the rules. Among them, the performance of the proposed Temporal Weighted Utility Association Rule Mining (TWUARM) using sliding window processing model was found to be most relevant as discussed in section 5.3. During the literature review, it was very clear that state-of-the-art techniques are not sufficient to extract meaningful and useful association rules. Hence, the existing Weighted Utility Association Rule Mining (WUARM) discussed in paper [36] (Dhanda, 2011) was revitalized as TWUARM, and tuned to extract utility rules from the Human Rights domain. The following pseudo-code explains the working of TWUARM in brief. The algorithm works similar to the exiting Apriori Algorithm [29] (Jiawei & Micheline, 2006) till step 1(b) except for the processing model. That is, sliding window since the existence of data is temporal. Then in step 1(c), it extracts utility rules from the derived association rules based on the weight threshold. Then, temporal rules were discovered from these utility rules.

<u>Algorithm</u> : Temporal Weighted Utility Association Rule Mining (TWUARM)

<u>Inputs</u> : Refined Dataset, min_support=65%, min_confidence = 85%,
wt_threshold = 20.0, win_size = 100, win_slide_ratio = win_size / 3

<u>Outputs</u> : Temporal Association Rules in each window

1. Scan the 100 tuples (= win_size) from the Refined dataset.
   (a) Extract frequent item-sets from the window. (With respect to min_support)
   (b) Identify association rules from frequent item-sets. (With respect to min_confidence).
   (c) Filter out the utility rules from the association rules. (With respect to wt_threshold)
   (d) Temporal rules were discovered from utility rules.
   (e) Slide the window (with respect to win_slide_ratio)
2. Repeat Step 1 until all the tuples is scanned at-least once.

No	Sub-Domains	Relevance Weight-age, R_i
1	Killed	150
2	Harassment	130
3	Kidnapped	130
4	Criminal	110
5	Protester	110
6	Rescued	100
7	Rehabilitation	100
8	Juvenile Home	100
9	Arrested	80
10	Law	80
11	Women	70
12	Child	70
13	Human	70

TABLE 2 : RELEVANCE WEIGHTAGE ASSIGNED TO 13 SUB-DOMAINS

Justification for finalizing parameter values for the TWUARM algorithm in Association Rule Mining component of CDME is as follows: (1) Window_size set to 100, under the assumption that #tweets/day shall not exceed 100; (2) Min_support and min_confidence was set to 65% and 85%, after experimental study and; (3) Weight_Threshold was set to 20.0, to confine the #Rules generated to a specific limit. Similar to the profit-weight assigned to items in the transaction dataset discussed in paper [36] (Dhanda, 2011), relevance-weight was given to the 13 sub-domains according to their impact upon the topic of discussion in the tweet as shown in Table 2.

While assigning weight-age for the sub-domains, window size and number of sub-domains were the two parameters considered. So, the total relevance weight, $\sum R_i$ was set to 1300 (window size * #sub-domains). This total weight was divided among the 13 sub-domains with respect to their impact in the Human Rights domain as shown in Table 2. For example, sub-domain 'killed' has more impact over the viewer than the sub-domain 'arrested'.

The profit-ratio and profit-weight formulas were re-defined for the Human Rights domain as follows. Refer Appendix C, for the formulae to calculate Profit-Ratio and Profit-Weight defined in paper [36] (Dhanda, 2011). Given an itemset $X = \{i_1, i_2 . . . i_n\}$,

Relevance ratio, $\text{Q-factor}_i = \dfrac{R_i}{\sum R_i}$,where, R_i is the relevance of item i. … (1)

Relevance-Weight, $\text{RW-factor}(X) = \sum \text{freq}(X) * \text{Q-factor}_i$ …(2)

3.5 Data Visualization

Finally to for quick inference, locations related to association rules were mapped onto Google Maps as heat maps via JavaScript API [49,50] Various Python packages were tried to plot locaions on the Google Maps, among them gmaps proved to be effecient in plotting Heat Maps.

3.6 Experimental Setup

The experiment was performed on Intel® Core™ i3-2350M CPU @ 2.30GHz × 4 with 8GB main memory running on Ubuntu 16.04 LTS. All programs were coded in Python using Stani's Python Editor (SPE 0.8.4.h). (Michiels, 2010). An application named "Mining Concept Drift" was created in Twitter for generating consumer keys and access tokens. These were then, used to extract real-time user timeline tweets from Shakti Vahini (@shaktivahini) and Human Rights Watch (@hrw) Twitter accounts.

Tweets were extracted using tweepy Python module downloaded from GitHub. The latest 3200 tweets were downloaded on a day-to-day basis. These tweets were then merged, to extract distinct tweets to construct the final raw dataset. While downloading the tweets, tiny URLs in the tweets were expanded and their corresponding web pages were crawled simultaneously.

For extracting, expanding and crawling the downloaded tweets, Python modules such as urllib, urlclean and Beautiful Soup were used. Upon these three tweet components, cleansing and grouping were performed to derive entries into the merged Tweet Corpus. For cleansing the tweet components, Python modules such as urlclean, Beautiful Soup, urllib, WordNet Lemmatizer, Newspaper, PyEnchant, PyNER, gmaps, pandas, NLTK were used. Finally, temporal association rules were mined from the refined dataset using the Sliding Window Processing Model.

Chapter – 4 : RESULTS AND DISCUSSIONS

As mentioned in Chapter 3, the entire work was performed in two phases. Even though each one of them could be performed individually, the methodology adopted for this experimental study, was sequential. In the first phase, the temporal rules was derived. In the next step, location information was mapped upon to those temporal rules for extracting inferences. Chapter 5 is also divided into three sections; with the Section 4.1 detailing the normalized dataset extracted for location based temporal rule mining. Section 4.2 mainly focus on the proof-of-concept for components included in the proposed Location based Alert System.

4.1 Data-set

Two distinct datasets were used for the experiments. For the first phase, dataset was created from he user-timeline tweets downloaded from @shaktivahini to extract location based temporal rules is as follows: Table 3 : Sample Dataset for extracting temporal rules

Tweet id	Created at	Locations in tweets	Domain specific keywords
888723904382083073	2017-07-22 11:34:25	west bengal west bengal west bengal kolkata kolkata delhi jharkhand guwahati kolkata ranchi	child law harassment woman protester rescued arrested juvenile criminal kidnapped killed human
888721552748855297	2017-07-22 11:25:04	sonipat	woman harassment law protester child rescued juvenile killed human
888711948744785920	2017-07-22 10:46:54	mahaguna moderne noida delhi mahaguna moderne west bengal mahaguna moderne noida mahaguna moderne noida delhi	harassment protester woman rehabilitation child human arrested law criminal rescued kidnapped killed
888321262329704448	2017-07-21 08:54:27	kolkata west bengal kolkata kolkata jharkhand guwahati kolkata ranchi	criminal harassment arrested rehabilitation kidnapped rescued law woman child protester
....	.....	....	....

The dataset for second phase, was created from first phase dataset by deriving the frequency of locations to plot them on Google Maps. Table 4 : Sample dataset for visualization is as follows:

Location	Latitude	Longitude	Frequency
adyar India	13.0011774	80.2564957	10
agartala India	23.831457	91.2867777	27
agra India	27.1766701	78.0080745	6
akkalkuva India	21.5546245	74.0159499	43
....	...	...	....

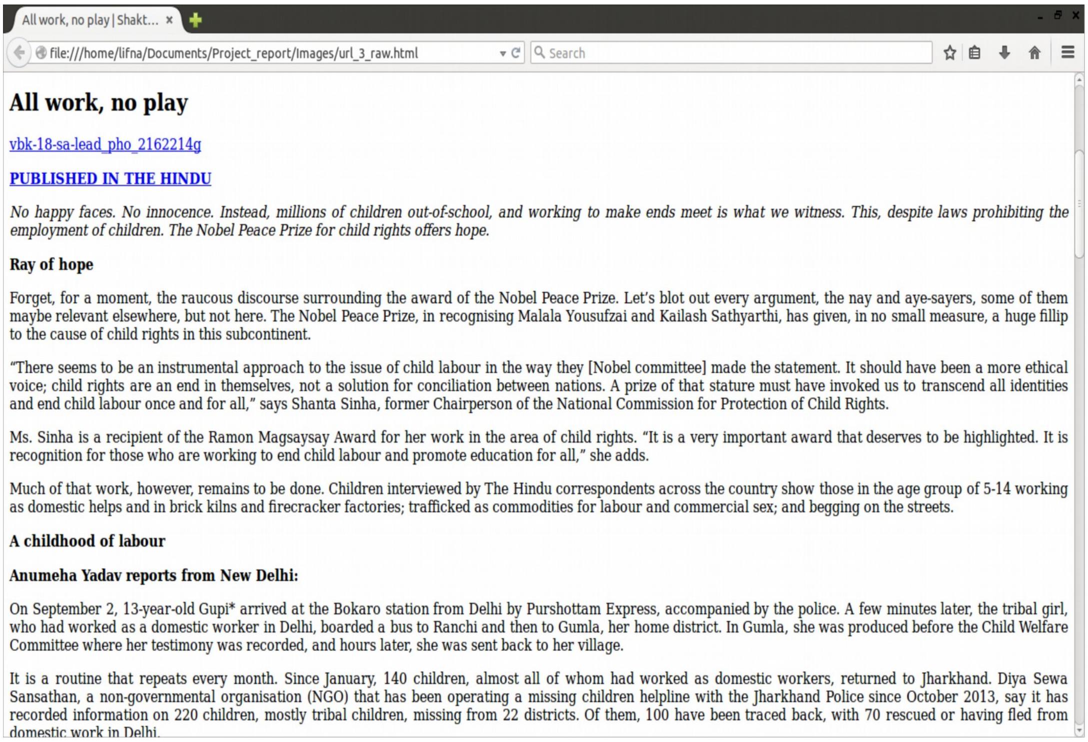
4.2 Proof of Concept

Appendix D and Appendix E are the results obtained after cleansing the first two tweet components; original tweet text and expanded tiny-URL respectively. The cleansed stem words extracted from each component is listed as the second last column in both the tables. Also, for quick reference, the number of stem words, in the respective cleansed tweet component is mentioned as the last column of Appendix D and Appendix E.

In some extreme cases, only the third tweet component proved to render some valid stem words relevant to the Human Rights domain. For example, in Tweet #3, there does not exist any stem words associated with Human Rights domain even after expanding the tiny-URL present in it. This is visible from the Appendix D and Appendix E. Here comes the relevance of crawling the expanded tinyURL. Table 5, illustrates the data gathering procedure associated with Tweet #3.

No	Methods	Associated Outputs
#1	Downloaded Tweet	All work, no play http://t.co/vBFJ9lD8TFhttp://t.co/xJuyK95kwu.
#2	Expanded tiny URL in Tweet	All work no play, http://shaktivahini.org/shakti-vahini-2/all-work-no-play-2, https://twitter.com/SHAKTIVAHINI/status/524217442977853440/photo/1.
#3	Crawled webpage of the Expanded tinyURL	Refer Figure 3

TABLE 5 : ILLUSTRATES THE ENTIRE DATA GATHERING PROCEDURE ASSOCIATED WITH TWEET#3


All work, no play | Shakt...
file:///home/lifna/Documents/Project_report/Images/url_3_raw.html

All work, no play

vbk-18-sa-lead pho 2162214g

PUBLISHED IN THE HINDU

No happy faces. No innocence. Instead, millions of children out-of-school, and working to make ends meet is what we witness. This, despite laws prohibiting the employment of children. The Nobel Peace Prize for child rights offers hope.

Ray of hope

Forget, for a moment, the raucous discourse surrounding the award of the Nobel Peace Prize. Let's blot out every argument, the nay and aye-sayers, some of them maybe relevant elsewhere, but not here. The Nobel Peace Prize, in recognising Malala Yousufzai and Kailash Sathyarthi, has given, in no small measure, a huge fillip to the cause of child rights in this subcontinent.

"There seems to be an instrumental approach to the issue of child labour in the way they [Nobel committee] made the statement. It should have been a more ethical voice; child rights are an end in themselves, not a solution for conciliation between nations. A prize of that stature must have invoked us to transcend all identities and end child labour once and for all," says Shanta Sinha, former Chairperson of the National Commission for Protection of Child Rights.

Ms. Sinha is a recipient of the Ramon Magsaysay Award for her work in the area of child rights. "It is a very important award that deserves to be highlighted. It is recognition for those who are working to end child labour and promote education for all," she adds.

Much of that work, however, remains to be done. Children interviewed by The Hindu correspondents across the country show those in the age group of 5-14 working as domestic helps and in brick kilns and firecracker factories; trafficked as commodities for labour and commercial sex; and begging on the streets.

A childhood of labour

Anumeha Yadav reports from New Delhi:

On September 2, 13-year-old Gupi* arrived at the Bokaro station from Delhi by Purshottam Express, accompanied by the police. A few minutes later, the tribal girl, who had worked as a domestic worker in Delhi, boarded a bus to Ranchi and then to Gumla, her home district. In Gumla, she was produced before the Child Welfare Committee where her testimony was recorded, and hours later, she was sent back to her village.

It is a routine that repeats every month. Since January, 140 children, almost all of whom had worked as domestic workers, returned to Jharkhand. Diya Sewa Sansathan, a non-governmental organisation (NGO) that has been operating a missing children helpline with the Jharkhand Police since October 2013, say it has recorded information on 220 children, mostly tribal children, missing from 22 districts. Of them, 100 have been traced back, with 70 rescued or having fled from domestic work in Delhi.

Figure 3. Snapshot of crawled webpage associated with the expanded tiny-URL present in Tweet#3.

From Appendix G, it was clear that it is difficult to extract meaningful rules from the ocean of stem words relevant to the Human Rights domain. So related words were identified and grouped under the 13 sub-domains as depicted in Table 6 and the resulting sample refined dataset is shown in Appendix H.

No	Sub-domains	Sample Stem words under sub-domain
1	Killed	Assault, murder, murdered, killing, ...
2	Harassment	Torture, tortured, prostitution, ...
3	Kidnapped	Kidnap, sold, nabbed, stolen, ...
4	Criminal	Racketeer, racket, kingpin, ...
5	Protester	Violence, activist, activists, reform, …
6	Rescued	Rescue, rescues, saves, saved, …
7	Rehabilitation	secured, securing, help, rehabilitate, …
8	Juvenile Home	Juvenile, juveniles, …
9	Arrested	Arrest, arrested, arresting, arrests, ...
10	Law	Policymakers, law, laws, legal, . . .
11	Women	Maid, Waitress, mother, bride, . . .
12	Child	Children, children's, childhood, . . .
13	Human	Human, humane, humans, . . .

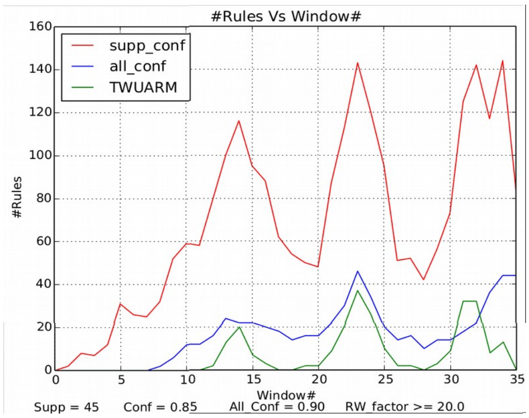
TABLE 6 : SAMPLE GROUPING TABLE USED FOR GROUPING STEMWORDS INTO 13 SUB-DOMAINS

To finalize upon the Association Rule Mining technique, the following three methods listed in Table 7 were applied and experimented with varying the parameter values for comparison using Sliding Window Processing Model.

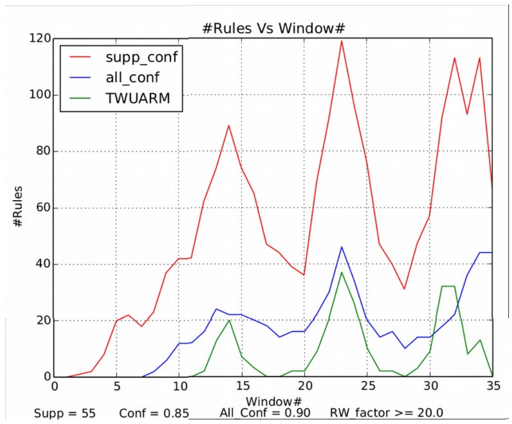
No	Method	Parameters
1	Support-Confidence Framework	Support, Confidence
2	Augment Support Confidence Framework with correlation measure [29] (Jiawei & Micheline, 2006)	Support, Confidence, All_confidence
3	Temporal Weighted Utility Association Rule Mining (TWUARM)	Support, Confidence, Weight_threshold

TABLE 7 : ASSOCIATION RULE MINING TECHNIQUES EXPERIMENTED UPON THE REFINED DATASET

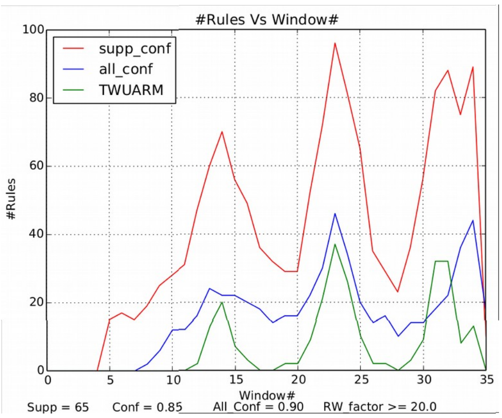
To compare the selected mining techniques, the parameter values were set as follows: (1) min_confidence = 85%; (2) all_confidence = .90; (3) weight_threshold = 2.0; (4) window_size = 100; and support was varied between 45%, 55%, ..., 95%. Appendix I, Appendix J and Appendix K gives a gist of the number of rules generated by the three mining techniques when support = 65%. For quick reference, the number of rules generated by these mining techniques in each window was then merged for analysis and depicted in Appendix L. From the consolidated chart in Appendix L, it was evident that, there was a steep decline in the number of rules and the refined rules were of great significance in the selected domain of Human Rights. This is clearly visible from the windows 12, 16, 21 and 26.



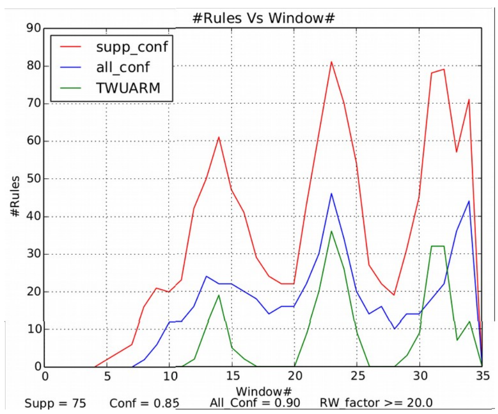
(a)



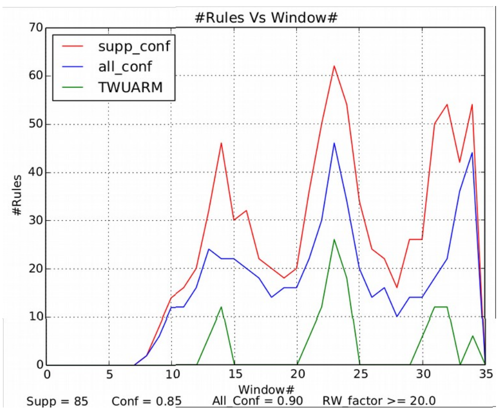
(b)



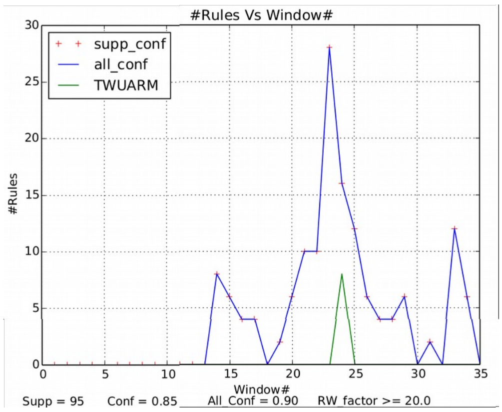
(c)



(d)



(e)



(f)

Figure 4. Graph plotted for comparing SC Framework, Augmented SC Framework with all_confidence and TWUARM to varying support. (a) Support = 45; (b) Support = 55; (c) Support = 65; (d) Support = 75; (e) Support = 85 and (f) Support = 95.

For quick visualization of the experimental results, consolidated graphs were plotted as shown in Figure 4 based on the chart in Appendix L. Even-though the peak in the graphs indicates a hike in the number of rules generated, indirectly it conveys that that the Human Rights domain was vibrant at that particular window. Comparing the results obtained from the three methods, TWUARM derived meaningful and useful association rules. This proves the efficiency and accuracy of TWUARM over the previous methods.

From the graphs plotted, parameters for TWUARM were set with the following values; (1) support = 65%, (2) confidence = 85% and (3) wt_threshold = 20.0. The support value was selected from graph 4(c), after which the number of TWUARM rules showed a gradual decline. The rules were null in some windows beyond this point. The above values were empirically standardized for the system. A detailed timeline chart is included as Appendix M which lists the temporal rules in their windows. After justifying the Association Rule Mining technique, it's now time to discover temporal rules. For this, consider the utility rules generated by TWUARM in windows 19, 20 and 21 at support 65%:

Window 19 : harassment, law --> juvenile , killed, juvenile --> protester

Window 20 : harassment, law --> juvenile, killed, juvenile --> protester
rehabilitation, juvenile --> harassment, harassment, killed --> protestor

Window 21 : killed, juvenile --> protester, rehabilitation, juvenile --> harassment.

There six events been reported in these windows. They are: Event A - harassment; Event B - law; Event C - killed; Event D - juvenile; Event E - protester; Event F - rehabilitation. When the rules are closely examined the temporal rules in these tweet windows were evident. Also, their transitive property is clearly visible in this illustration.

The location information gathered during the first phase was utilized to plot the following heat map depicted in Figure 5. From the heatmap, the following concusions can be made. Firstly, as per the Tweets from @shaktivahini, New Delhi is the the location which is more into discussion during the period as it is more into red compared to the other locations depicted in yellow (for locations such as Mumbai, Assam and Benguluru) and in green (for locations in Goa, Kerala, Tamil Nadu, Karanataka). Secondly, to extract precise and all over information there is need to integrate Social Analytics with News Analytics before integrating the developed system into eGovernance. Thus, the existence of transitive property has a valuable derivative inferences not only to predict events in the Human Rights domain, but also to serve as a Location based Alert System for concerned Government Departments for a smooth and effective e-Governance.

Figure 5. Location based heatmap over Indian Terrain for Human Rights violation incidents

Chapter – 5 : CONCLUSIONS AND FUTURE SCOPE

Section 5.1 details the findings and contributions derived from the above study. The extensions identified during the experimental study and the future scope in the Human Rights domains is discussed in Section 5.2.

5.1 Conclusion

The experiment started off with, downloading the user timeline tweets from @shaktivahini and @hrw in the Human Rights domain. To address the intrinsic properties of tweets; the data gathering, cleansing and grouping steps were performed extensively to derive refined dataset. Thus, at this stage, the first objective and the major contribution mentioned in the paper, i.e., addressing hidden context challenge has been achieved.

Since, sliding window processing model was efficient as mentioned in paper [51] (Lifna & Vijayalakshmi, 2015), the refined dataset was processed using the same model. For each window, association rules were mined and refined using TWUARM. The interesting patterns of temporal association between the rules were noticed during the study. Thus, the second major contribution was revitalizing WUARM into TWUARM for such a vibrant data stream from Twitter. The concluding para of Section 4.2, distinctly states how the study was able to attain its second objective, i.e., to discover the existence of temporal rules. The third objective was achieved by providing a heatmap over Indian terrain based on the Human Rights violation incidents extracted from tweets.

5.2 Future Scope

As a future scope we are planning to extend the work towards the goal of integrating the proposed system with eGovernance. Towards the same the following subgoals have been identified.

1. Creation of Ontology based domain dictionary for identifying the concepts.
2. Integrate News Analytics with Social Analytics for gathering the exact scenario.
3. Real time prediction of events.
4. Notify the Government bodies regarding predict events with probability.
5. Improve system performance by incorporating Big Data Tools.

APPENDIX

A. Raw user timeline tweets downloaded from @shaktivahini

Tweet_id	Created_at	RT	Tweet_text
5242797375 73573000	2014-10-20 19:23:24	0	Have you heard 'INTERVIEW OF ARADHANA SINGH INSPECTOR ANTI HUMAN TRAFFICKING UNIT JHARKHAND' on #SoundCloud? https://t.co/VaTJwXNRtp
5242792246 60529000	2014-10-20 19:21:22	1	Have you heard 'A Police Officer Can be the World To A Trafficked Victim' by Shakti Vahini on #SoundCloud? https://t.co/m8okEbctIj
5242174429 77853000	2014-10-20 15:15:52	1	"All work, no play http://t.co/vBFJ9lD8TF http://t.co/xJuyK95kwu"
5241355102 73323000	2014-10-20 09:50:18	0	Global #ModernSlavery Directory profiles NGOs fighting to #endslavery. http://t.co/HMMzQRFmty #shaktivahini http://t.co/RZugjLqxFn
…	…	..	…

B. Raw user timeline tweets downloaded from @hrw

Tweet_id	Created_at	RT	Tweet_text
5824919724 40616960	2015-03-30 10:38:03	71	RT @KenRoth: Help @HRW reverse Obama's cowardly refusal to prosecute Bush torturers. Sign petition and RT: http://t.co/G1Kf8EkmOF http://t.…
5824918586 32368128	2015-03-30 10:37:35	30	RT @belkiswille: #Kuwait should respect people's freedom to demonstrate peacefully against the government http://t.co/vqdMuOaZfU
5824917194 35964417	2015-03-30 10:37:02	7	RT @WenzelMichalski: Kyrgyzstan is reaching out to Europe – but regarding rights issues it's inching closer to Russia. @HughAWilliamson htt…
5824913661 95884032	2015-03-30 10:35:38	24	RT @snorthfield45: When 'religious freedom' means the right to discriminate. @hrw's @Graemecreid on Indiana's odious Bill 101 #LGBT http://…
…	…	..	…

C. Formula to calculate Profit-Ratio & Profit-Wt. [36] (Dhanda, 2011)

Given an itemset X = {i1, i2. . . in },

$$\text{Profit Ratio, Q-factori} = \frac{Pi}{\sum Pi}$$ where, Pi is the profit on an item I.

$$\text{Profit-Weight, PW-factor}(X) = \sum \text{frequency}(X) * \text{Q-factori}$$

D. Original Tweet Text component and its corresponding cleansed Tweet Text

Tweet #	Tweet_id	Original Tweet_text	Cleansed Tweet_text	#Stem words
#1	524279737 573573000	Have you heard 'INTERVIEW OF ARADHANA SINGH INSPECTOR ANTI HUMAN TRAFFICKING UNIT JHARKHAND' on #SoundCloud? https://t.co/VaTJwXNRtp	Human, trafficking	2
#2	524279224 660529000	Have you heard 'A Police Officer Can be the World To A Trafficked Victim' by Shakti Vahini on #SoundCloud? https://t.co/m8okEbctIj	Police, trafficked, victim	3
#3	524217442 977853000	"All work, no play http://t.co/vBFJ9lD8TF http://t.co/xJuyK95kwu"	-	-
#4	524135510 273323000	Global #ModernSlavery Directory profiles NGOs fighting to #endslavery. http://t.co/HMMzQRFmty #shaktivahini http://t.co/RZugjLqxFn	-	-
#5	524106503 599309000	PANNALAL TRAFFICKER FROM JHARKHAND WHO WAS ABSCONDING ARRESTED BY JHARKHAND POLICE IN NEW DELHI #shaktivahini http://t.co/pe0FheSrkB	Trafficker, absconding, arrested	3
	. . .	. . .	. . .	. . .
#74	518485959 025766000	GUIDELINES & PROTOCOLS Medico-legal care for survivors/victims of Sexual Violence http://t.co/wGHbMAE9Fq	Guidelines, protocols, survivors, victims, sexual, violence	6
. . .	. . .	. . .	. . .	. . .

E. Tiny URL, its expanded form and its corresponding cleansed expanded URL.

Tweet #	tinyURL	Expanded tinyURL	Cleansed expanded URL	#Stem words
#1	https://t.co/VaTJwXNRtp	https://soundcloud.com/shakti-vahini/interview-of-aradhana-singh-anti-human-trafficking-unit-jharkhand?utm_source=soundcloud&utm_campaign=share&utm_medium=twitter	Human, trafficking	2
#2	https://t.co/m8okEbctIj,	https://soundcloud.com/shakti-vahini/a-police-officer-can-be-the?utm_source=soundcloud&utm_campaign=share&utm_medium=twitter	police	1
#3	http://t.co/vBFJ9lD8TF, http://t.co/xJuyK95kwu	http://shaktivahini.org/shakti-vahini-2/all-work-no-play-2, https://twitter.com/SHAKTIVAHINI/status/524217442977853440	-	-
	. . .	. . .	. . .	. . .
#74	http://t.co/wGHbMAE9Fq	http://nlrd.org/resources-womens-rights/rape-laws/government-notifications-advisories-rape-laws/guidelines-protocols-medico-legal-care-for-survivorsvictims-of-sexual-violence,	Resources, womens, rights, rape, laws, care, governement, sexual, notifications, advisories, guidelines, protocols, legal, medico, violence	15
. . .	. . .	. . .	. . .	. . .

F. Original Tweet along with its cleansed Tweet components before merging

Tweet #	Original Tweet	Cleansed		
		Tweet Text	Expanded URL	Crawled webpage
#1	Have you heard 'INTERVIEW OF ARADHANA INSPECTOR ANTI HUMAN TRAFFICKING UNIT JHARKHAND' on #SoundCloud? https://t.co/VaTJwXNRtp	Human, trafficking	Human, trafficking	trafficking
#2	Have you heard 'A Police Officer Can be the World To A Trafficked Victim' by Shakti Vahini on #SoundCloud? https://t.co/m8okEbctIj	Police, trafficked, victim	police	Trafficked, victim
#3	"All work, no play http://t.co/vBFJ9lD8TF http://t.co/xJuyK95kwu"	-	-	Children, witness, laws, prohibiting, child, argument, conciliation, invoked, maid, protection, trafficked, sex, begging, girls childhood, girl, rescued, fled, maternal, grandmother, stolen, sold, delineating, women, rehabilitating, patrol, flee, raped, prohibited, abusive, mother, sister, begged, addicted, abused, adolescent, prostitution, poverty, illegal, survive, trafficking, flees, kingpin, confesses, crime, bride, trafficker, arrested, nephew, slammed, assault, victim, violence, discover
. . .	. . .	. . .	. . .	. . .
#74	GUIDELINES & PROTOCOLS Medico-legal care for survivors/victims of Sexual Violence http://t.co/wGHbMAE9Fq	Guidelines, protocols, survivors, victims, sexual, violence	Resources, womens, rights, rape, laws, care, government, sexual, notifications, advisories, guidelines, protocols, legal, medico, violence	Survivors, sexual, supreme, court, legislations, laws, women, advisories, violence, womens, child, protection, children, juvenile, trafficking, sex, rape, dowry, attacks, killing, crimes, female, harassment, compact, pharmacy, approaching, treating, documenting, girls, fulfilled, pills, guideline, plugged, quarters, improvements, effected, treated, iterative, mitigating, holistic, cutting, boundaries, doctors, assault, course, hospital, denial, cognizable, punishable, fines, stringent, tolerance, offenders, sensitive, humane, attending, aspects, prosecution, experienced, documented, jurisdiction, incumbent, gratis, avail, appealing, contributing, practitioners, informed, request, improvement, prostitution, regulation, amendments, prohibition, victim, juveniles,
. . .	. . .	. . .	. . .	. . .

G. Single Merged Tweet Corpus

Tweet #	Merged Tweet Corpus	#Stem Words
#1	Human, trafficking	2
#2	Police, trafficked, victim	3
#3	Children, witness, laws, prohibiting, child, argument, conciliation, invoked, maid, protection, trafficked, sex, begging, girls childhood, girl, rescued, fled, maternal, grandmother, stolen, sold, delineating, women, rehabilitating, patrol, flee, raped, prohibited, abusive, mother, sister, begged, addicted, abused, adolescent, prostitution, poverty, illegal, survive, trafficking, flees, kingpin, confesses, crime, bride, trafficker, arrested, nephew, slammed, assault, victim, violence, discover	54
. . .	. . .	. . .
#74	Survivors, sexual, supreme, court, legislators, laws, women, advisories, violence, womens, child, protection, children, juvenile, trafficking, sex, rape, dowry, attacks, killing, crimes, female, harassment, pharmacy, approaching, treating, documenting, girls, guideline, plugged, quarters, improvements, effected, treated, iterative, mitigating, holistic, cutting, boundaries, doctors, assault, hospital, denial, cognizable, punishable, fines, stringent, tolerance, offenders, sensitive, humane, attending, aspects, prosecution, experienced, documented, jurisdiction, incumbent, gratis, avail, appealing, contributing, practitioners, informed, request, improvement, prostitution, regulation, amendments, prohibition, victim, juveniles,	73
. . .	. . .	. . .

H. Sample Refined Data-set

Tweet #	Concepts	#Concepts
#1	Human, harassment	2
#2	Law, harassment, human	3
#3	Child, protester, law, rehabilitation, women, harassment, rescued, kidnapped, criminal, arrested, killed, human	12
. . .	. . .	. . .
#74	Child, killed, harassment, criminal, protester, rescued, rehabilitation, juvenile, law, women, human	11
. . .	. . .	. . .

I. Rules generated in each window for SC Framework

(Confidence = 85% and Support = 70%)

Window #	Rules Generated			#Rules
#1	-			0
. . .	. . .			. . .
#5	-			0
#6	{criminal} -> {harassment} {human} -> {harassment} {arrested} -> {harassment}, {arrested} -> {child}	{killed} -> {protester}, {protester} -> {killed} {killed} -> {child}	{rehabilitation} -> {harassment} {killed} -> {harassment} {harassment} -> {human}	10
#7	{arrested} -> {harassment} {criminal} -> {harassment} {human} -> {harassment}	{arrested} -> {child}, {criminal} -> {child} {killed} -> {protester}	{rehabilitation} -> {harassment} {killed} -> {harassment} {harassment} -> {human}	9
#8	{juvenile} -> {law}, {juvenile} -> {child} {human} -> {harassment} {juvenile} -> {protester} {arrested} -> {harassment} {criminal} -> {harassment}	{juvenile} -> {woman}, {human} -> {child} {harassment} -> {human} {human} -> {protester}, {rescued} -> {child} {killed} -> {harassment}	{rehabilitation} -> {harassment} {juvenile} -> {rescued}, {human} -> {woman} {killed} -> {protester}, {child} -> {human} {law} -> {juvenile}	18
. . .	. . .			. . .

J. Rules generated on each window for Augmented Support Confidence Framework

(Confidence = 85%, Support = 70% and All_confidence = 0.90)

Window #	Rules Generated			#Rules
#1	-			0
#2	-			0
. . .	. . .			. . .
#7	-			0
#8	{human} -> {harassment}		{harassment} -> {human}	2
#9	{juvenile} -> {law} {juvenile} -> {protester}	{human} -> {harassment} {law} -> {juvenile}	{protester} -> {juvenile} {harassment} -> {human}	6
#10	{juvenile} -> {law} {juvenile} -> {protester} {juvenile} -> {rescued} {juvenile} -> {woman}	{law} -> {juvenile} {protester} -> {juvenile} {rescued} -> {juvenile} {child} -> {juvenile}	{human} -> {harassment} {juvenile} -> {child} {woman} -> {juvenile} {harassment} -> {human}	12
#11	{juvenile} -> {law} {juvenile} -> {protester} {juvenile} -> {woman} {law} -> {juvenile}	{protester} -> {juvenile} {woman} -> {juvenile} {protester} -> {killed} {rescued} -> {juvenile}	{juvenile} -> {child} {child} -> {juvenile} {killed} -> {protester} {juvenile} -> {rescued}	12
#12	{juvenile} -> {child} {juvenile} -> {protester} {juvenile} -> {woman} {woman} -> {juvenile} {protester} -> {juvenile} {juvenile} -> {law}	{law} -> {juvenile} {child} -> {juvenile} {juvenile} -> {harassment} {killed} -> {protester} {child} -> {human}	{human} -> {harassment} {protester} -> {killed} {harassment} -> {human} {harassment} -> {juvenile} {human} -> {child}	16
#13	{juvenile} -> {child} {juvenile} -> {harassment} {juvenile} -> {protester} {law} -> {juvenile} {protester} -> {juvenile} {killed} -> {protester} {human} -> {child} {child} -> {human}	{child} -> {juvenile} {human} -> {harassment} {juvenile} -> {law} {protester} -> {killed} {juvenile} -> {killed} {killed} -> {juvenile} {harassment} -> {juvenile} {juvenile} -> {woman}	{harassment} -> {human} {woman} -> {juvenile} {juvenile} -> {protester, killed} {killed} -> {protester, juvenile} {protester, juvenile} -> {killed} {protester} -> {killed, juvenile} {protester, killed} -> {juvenile} {killed, juvenile} -> {protester}	24
. . .	. . .			. . .

K. Rules generated on each window for TWUARM

(Confidence = 85%, Support = 70% and Wt_Threshold = 20.0)

Window #	Rules Generated		#Rules
#1	-		0
. . .	. . .		. . .
#11	-		0
#12	{killed, juvenile} -> {protester}	{protester, killed} -> {juvenile}	2
#13	{harassment, law} -> {juvenile} {killed, juvenile} -> {protester} {protester, killed} -> {juvenile} {juvenile} -> {protester, killed} {killed} -> {protester, juvenile} {protester, juvenile} -> {killed}	{law} -> {harassment, juvenile} {law, juvenile} -> {harassment} {protester} -> {killed, juvenile} {harassment, juvenile} -> {law} {juvenile} -> {harassment, law}	11
#14	{harassment, law} -> {juvenile} {killed, juvenile} -> {protester} {rehabilitation, juvenile} -> {harassment} {harassment, killed} -> {protester} {harassment, protester} -> {killed} {law, juvenile} -> {harassment} {law} -> {harassment, juvenile} {harassment, rehabilitation} -> {juvenile} {protester, killed} -> {juvenile} {harassment, juvenile} -> {law}	{juvenile} -> {harassment, law} {protester, juvenile} -> {killed} {rehabilitation} -> {harassment, juvenile} {juvenile} -> {protester, killed} {killed} -> {protester, juvenile} {harassment} -> {law, juvenile} {protester} -> {killed, juvenile} {harassment, juvenile} -> {rehabilitation} {juvenile} -> {harassment, rehabilitation}	19
#15	{rehabilitation, juvenile} -> {harassment} {killed, juvenile} -> {protester} {harassment, rehabilitation} -> {juvenile}	{protester, killed} -> {juvenile} {protester, juvenile} -> {killed}	5
. . .	. . .		. . .

L. #Rules generated after applying the three Association rule Mining Techniques over Refined @shaktivahini tweets (#Concepts = 13, win_slide_ratio = 33 (win_size / 3), min_confidence = 85%, All_confidence = 90%, Wt_threshold = 20.0)

Win_size = 100			Support = 45			Support = 55			Support = 65			Support = 75			Support = 85			Support = 95		
window	win_start	win_end	SC	ASC	TWUARM	SC	ASC	TWUARM	SC	ASC	TWUARM	SC	ASC	TWUARM	SC	ASC	TWUARM	SC	ASC	TWUARM
0	0	99	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	33	132	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	66	165	8	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	99	198	7	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	132	231	12	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	165	264	31	0	0	20	0	0	15	0	0	2	0	0	0	0	0	0	0	0
6	198	297	26	0	0	22	0	0	17	0	0	4	0	0	0	0	0	0	0	0
7	231	330	25	0	0	18	0	0	15	0	0	6	0	0	0	0	0	0	0	0
8	264	363	32	2	0	23	2	0	19	2	0	16	2	0	2	2	0	0	0	0
9	297	396	52	6	0	37	6	0	25	6	0	21	6	0	8	6	0	0	0	0
10	330	429	59	12	0	42	12	0	28	12	0	20	12	0	14	12	0	0	0	0
11	363	462	58	12	0	42	12	0	31	12	0	23	12	0	16	12	0	0	0	0
12	396	495	79	16	2	62	16	2	**47**	**16**	**2**	42	16	2	20	16	0	0	0	0
13	429	528	100	24	13	74	24	13	60	24	13	50	24	11	32	24	6	0	0	0
14	462	561	116	22	20	89	22	20	70	22	20	61	22	19	46	22	12	8	8	0
15	495	594	95	22	7	74	22	7	56	22	7	47	22	5	30	22	0	6	6	0
16	528	627	88	20	3	65	20	3	**49**	**20**	**3**	41	20	2	32	20	0	4	4	0
17	561	660	62	18	0	47	18	0	36	18	0	29	18	0	22	18	0	4	4	0
18	594	693	54	14	0	44	14	0	32	14	0	24	14	0	20	14	0	0	0	0
19	627	726	50	16	2	39	16	2	29	16	2	22	16	0	18	16	0	2	2	0
20	660	759	48	16	2	36	16	2	29	16	4	22	16	0	20	16	0	6	6	0
21	693	792	87	22	9	69	22	9	**52**	**22**	**2**	43	22	9	36	22	6	10	10	0
22	726	825	113	30	21	92	30	21	72	30	21	62	30	20	50	30	12	10	10	0
23	759	858	143	46	37	119	46	37	96	46	37	81	46	36	62	46	26	28	28	8
24	792	891	120	34	26	96	34	26	81	34	26	70	34	26	54	34	18	16	16	0
25	825	924	95	20	10	76	20	10	65	20	10	54	20	9	34	20	0	12	12	0
26	858	957	51	14	2	47	14	2	**35**	**14**	**2**	27	14	0	24	14	0	6	6	0
27	891	990	52	16	2	40	16	2	29	16	2	22	16	0	22	16	0	4	4	0
28	924	1-23	42	10	0	31	10	0	23	10	0	19	10	0	16	10	0	4	4	0
29	957	1056	56	14	3	47	14	3	36	14	3	31	14	3	26	14	0	6	6	0
30	990	1089	73	14	9	57	14	9	56	14	9	45	14	9	26	14	6	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

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